

WHITE BAND ASSOCIATES

Cyber Security Internship Programme

ETHICAL HACKING — TASK 01

Information Gathering & Reconnaissance

Passive Reconnaissance Report

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Role	Cyber Security Intern
Target Assessed	Kristu Jayanti University (kristujayanti.edu.in)
Assessment Type	Passive Reconnaissance (OSINT)
Date of Assessment	26 June 2026
Report Classification	Internal — Internship Use Only

Objective

This task introduces the first phase of the Ethical Hacking methodology — Reconnaissance (Information Gathering). Every professional penetration test begins with the systematic collection of publicly available information about a target before any active testing is attempted. This report documents a purely passive reconnaissance exercise carried out against the official website of Kristu Jayanti University, using only publicly accessible OSINT tools. No system was accessed, scanned for vulnerabilities, or exploited at any stage of this exercise.

SCOPE NOTE All activity in this report is passive and non-intrusive. Only public WHOIS registries, DNS resolvers, browser-based technology fingerprinting tools, and publicly served files (robots.txt, sitemap.xml) were queried.

PART A

Target Selection

◆ Target Details

Website Name	Kristu Jayanti University (Deemed to be University)
URL	https://www.kristujayanti.edu.in/
Category	Higher Education Institution — Academic Website
Reason for Selection	Selected as the author's own college website, satisfying the task requirement of a legitimate, authorised, publicly accessible domain for passive OSINT practice.



Figure 1: Homepage of the target domain, kristujayanti.edu.in, viewed prior to beginning reconnaissance.

The target website, kristujayanti.edu.in, belongs to Kristu Jayanti University — a Deemed-to-be University in Bengaluru. It was selected as the reconnaissance target because it is a legitimate public domain that the author has authorisation to assess passively. The homepage confirms the site is live, publicly accessible, and hosts a rich academic portal with navigation menus, image sliders, and multiple sub-sections, indicating a moderately complex web application that warrants thorough OSINT investigation.

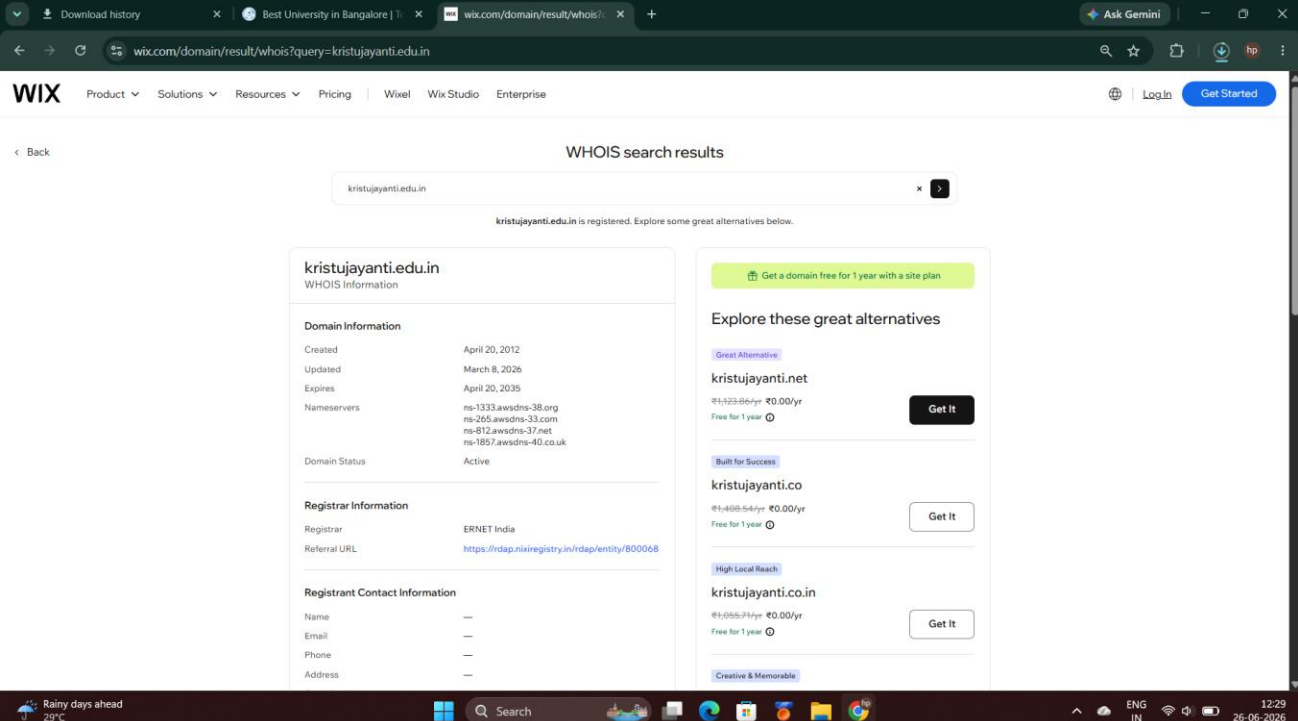
PART B

WHOIS Lookup

A WHOIS lookup was performed against `kristujayanti.edu.in` using the Wix WHOIS lookup service to retrieve domain registration metadata. WHOIS is a publicly accessible protocol that returns information about who registered a domain, through which registrar, and the creation, update, and expiry dates of the registration.

◆ WHOIS Findings

Domain Name	kristujayanti.edu.in
Registrar	ERNET India
Registration Date	20 April 2012
Last Updated	8 March 2026
Expiry Date	20 April 2035
Domain Status	Active
Name Servers	ns-1333.awsdns-38.org, ns-265.awsdns-33.com, ns-812.awsdns-37.net, ns-1857.awsdns-40.co.uk
Registrant Contact	Redacted — not publicly disclosed



The screenshot shows the Wix website's WHOIS search results for the domain `kristujayanti.edu.in`. The page displays the domain's registration details, including its creation date (April 20, 2012), last update (March 8, 2026), and expiry date (April 20, 2035). The registrar is listed as ERNET India, and the domain status is Active. The name servers are ns-1333.awsdns-38.org, ns-265.awsdns-33.com, ns-812.awsdns-37.net, and ns-1857.awsdns-40.co.uk. The registrant contact information is redacted. The page also suggests alternative domain names for purchase, such as `kristujayanti.net`, `kristujayanti.co`, and `kristujayanti.co.in`, each with a 'Get It' button.

Domain Information	
Created	April 20, 2012
Updated	March 8, 2026
Expires	April 20, 2035
Nameservers	ns-1333.awsdns-38.org ns-265.awsdns-33.com ns-812.awsdns-37.net ns-1857.awsdns-40.co.uk
Domain Status	Active

Registrar Information	
Registrar	ERNET India
Referral URL	https://rdap.nixiregistry.in/rdap/entity/800068

Registrant Contact Information	
Name	—
Email	—
Phone	—
Address	—

Explore these great alternatives

- Great Alternative**
`kristujayanti.net`
₹1,123.00/yr ₹0.00/yr
Free for 1 year [Get It](#)
- Built for Success**
`kristujayanti.co`
₹1,408.54/yr ₹0.00/yr
Free for 1 year [Get It](#)
- High Local Reach**
`kristujayanti.co.in`
₹1,055.24/yr ₹0.00/yr
Free for 1 year [Get It](#)
- Creative & Memorable**

Figure 2: WHOIS results showing domain creation, update, expiry dates, name servers, and registrar information.

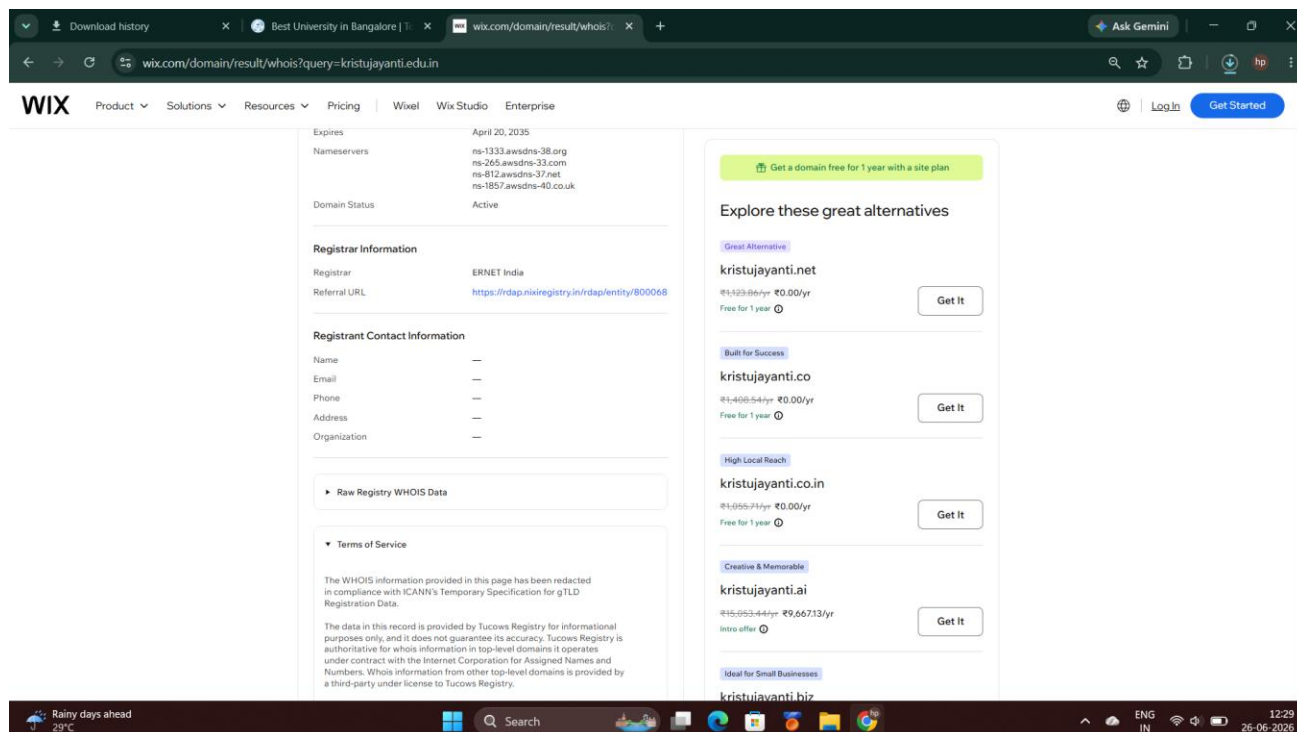


Figure 3: Continued WHOIS output confirming registrant contact fields are redacted under ICANN privacy policy.

The WHOIS results reveal that the domain has been registered since April 2012 — over a decade of established presence — and is active until 2035. All four name servers carry the awsdns-* suffix, confirming that DNS management has been fully delegated to Amazon Web Services Route 53 rather than the registrar's own infrastructure. Registrant contact details (name, email, phone, address) are redacted under ICANN's privacy specification, which is standard practice. The registrar, ERNET India, is the national education and research network responsible for .edu.in domains, confirming the university's legitimacy as a recognised higher education institution.

OBSERVATION DNS is fully managed via AWS Route 53. Registrant contact details are redacted under ICANN privacy policy. ERNET India as registrar confirms authorised .edu.in domain ownership.

PART C

DNS Enumeration

DNS records for `www.kristujayanti.edu.in` were enumerated using `DNSChecker.org`, querying against Google’s public DNS resolver (8.8.8.8). DNS stores multiple record types that collectively map a domain’s infrastructure — A records for web hosting, MX for mail routing, NS for name server delegation, and TXT for verification and anti-spoofing policies.

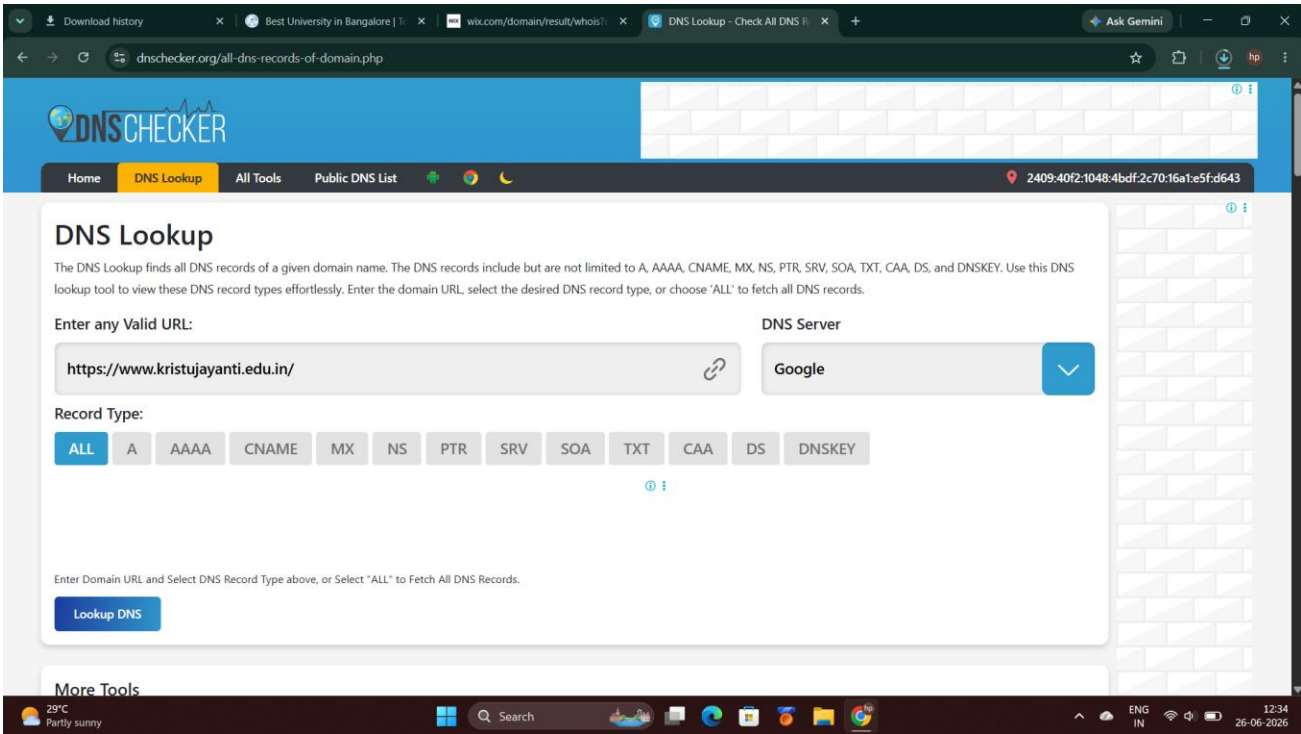


Figure 4: DNSChecker.org configured to query all record types for `www.kristujayanti.edu.in`.

◆ A Record (IPv4 Address)

Type	TTL	Address	Owner / ASN
A	15	13.235.5.4	Amazon Data Services India — AS16509
A	15	13.232.185.200	Amazon Data Services India — AS16509

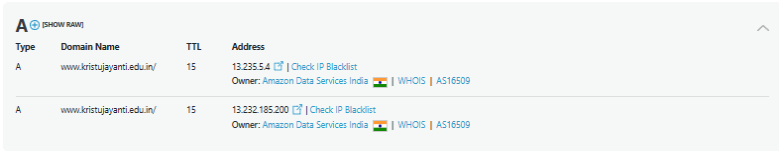
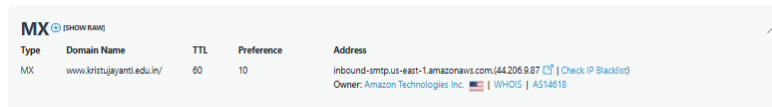


Figure 5: A record results — two IPv4 addresses resolving to Amazon Data Services India.

◆ MX Record (Mail Exchange)

Type	TTL	Preference	Address
MX	60	10	inbound-smtp.us-east-1.amazonaws.com (44.206.9.87)

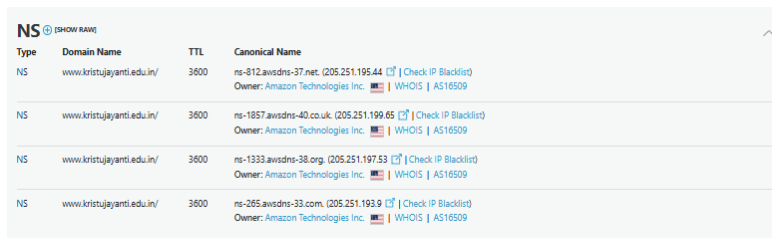


Type	Domain Name	TTL	Preference	Address
MX	www.kristujayanti.edu.in/	60	10	inbound-smtp.us-east-1.amazonaws.com (44.206.9.87) [Check IP Blacklist] Owner: Amazon Technologies Inc. [USA] WHOIS AS14618

Figure 6: MX record showing inbound mail routed through Amazon SES.

◆ NS Record (Name Servers)

Type	TTL	Name Server	Owner
NS	3600	ns-812.awsdns-37.net	Amazon Technologies — AS16509
NS	3600	ns-1857.awsdns-40.co.uk	Amazon Technologies — AS16509
NS	3600	ns-1333.awsdns-38.org	Amazon Technologies — AS16509
NS	3600	ns-265.awsdns-33.com	Amazon Technologies — AS16509



Type	Domain Name	TTL	Canonical Name
NS	www.kristujayanti.edu.in/	3600	ns-812.awsdns-37.net. (205.251.195.44) [Check IP Blacklist] Owner: Amazon Technologies Inc. [USA] WHOIS AS16509
NS	www.kristujayanti.edu.in/	3600	ns-1857.awsdns-40.co.uk. (205.251.199.65) [Check IP Blacklist] Owner: Amazon Technologies Inc. [USA] WHOIS AS16509
NS	www.kristujayanti.edu.in/	3600	ns-1333.awsdns-38.org. (205.251.197.53) [Check IP Blacklist] Owner: Amazon Technologies Inc. [USA] WHOIS AS16509
NS	www.kristujayanti.edu.in/	3600	ns-265.awsdns-33.com. (205.251.193.9) [Check IP Blacklist] Owner: Amazon Technologies Inc. [USA] WHOIS AS16509

Figure 7: NS records confirming all four authoritative name servers belong to Amazon Route 53.

◆ TXT Record

Type	TTL	Record Value
TXT	60	google-site-verification=VOrsWwWwEhAGFZkrnd4U1jeqPgWzlFyoBiXUFBGL3ig
TXT	60	google-site-verification=koWlUuRMXif5j3oqdug2npE7j5RF7yz7bUarZgEbLdil
TXT	60	v=spf1 include:_spf.google.com ~all

TXT	60	v=spf1 include:amazonses.com ~all
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Type	Domain Name	TTL	Record
TXT	www.kristujayanti.edu.in/	60	google-site-verification=V0r18WwBhAGF2Xrnd4U1jaeRgWdFycBXUFBGL3ig
TXT	www.kristujayanti.edu.in/	60	google-site-verification=ik2WuRvKXf53oqduq2npE7j5RF7yc7bUadGgblldli
TXT	www.kristujayanti.edu.in/	60	v=spf1 include:_spf.google.com ~all
TXT	www.kristujayanti.edu.in/	60	v=spf1 include:amazonses.com ~all

Figure 8: TXT records showing two Google site-verification tokens and two SPF anti-spoofing policies.

The DNS enumeration confirms that the university’s entire internet-facing infrastructure runs on AWS. The two A records point to Amazon Data Services India (Mumbai region), indicating load-balanced web hosting. Inbound email is handled by Amazon SES (us-east-1). All four NS records belong to Route 53, spanning multiple TLDs for resilience. The TXT records reveal two Google services integrated with the domain (likely Search Console and Workspace), and two SPF policies indicating that outgoing mail is sent through both Google Workspace and Amazon SES. The dual SPF configuration is normal but should be monitored, as a misconfigured SPF record could inadvertently allow spoofed email to pass spam filters.

OBSERVATION Entire infrastructure — web hosting, DNS, and inbound mail — is hosted on AWS. Two separate SPF policies indicate dual email-sending channels through Google Workspace and Amazon SES. Two Google site-verification tokens suggest multiple Google services are integrated.

PART D**Website Technology Identification**

The technology stack was fingerprinted using Wappalyzer, a browser extension that passively inspects HTTP response headers, HTML markup, and embedded scripts to identify frameworks, libraries, and server software — without sending any intrusive requests to the target.

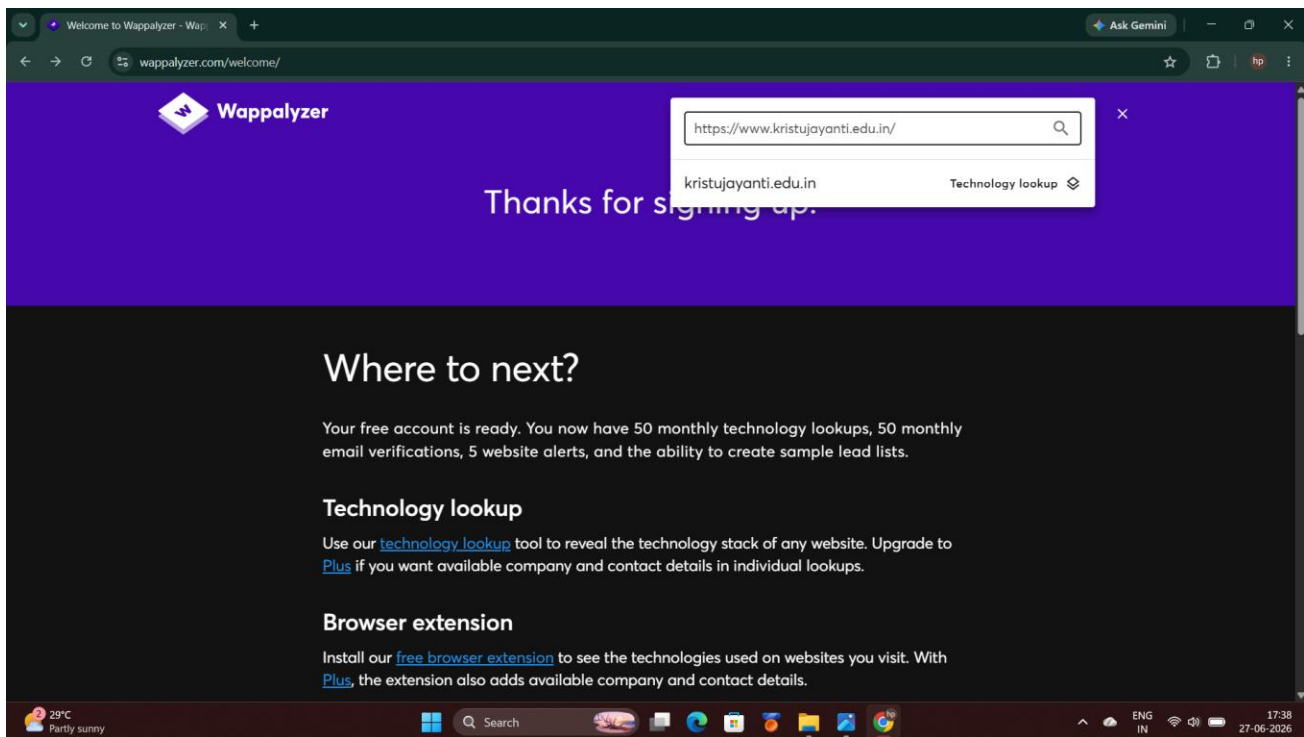


Figure 9: Wappalyzer queried against <https://www.kristujayanti.edu.in/>.

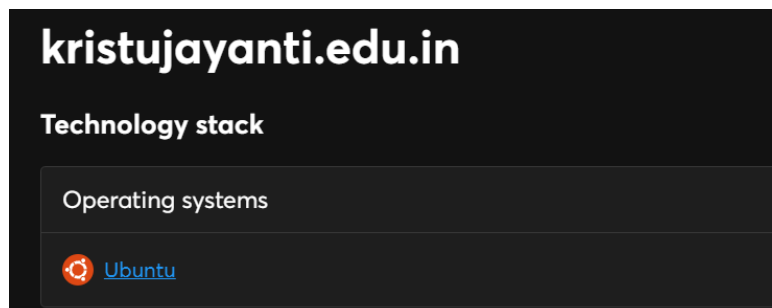
◆ Operating System

Figure 10: Server operating system identified as Ubuntu.

◆ UI Framework

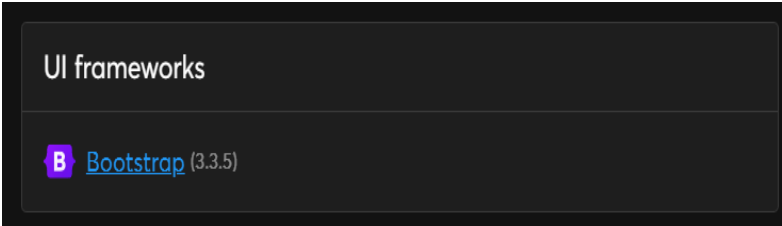


Figure 11: Front-end UI framework identified as Bootstrap version 3.3.5.

◆ Web Server

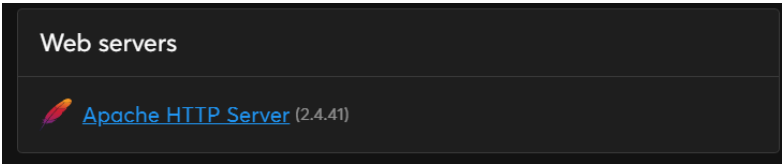


Figure 12: Web server software identified as Apache HTTP Server version 2.4.41.

◆ JavaScript Libraries

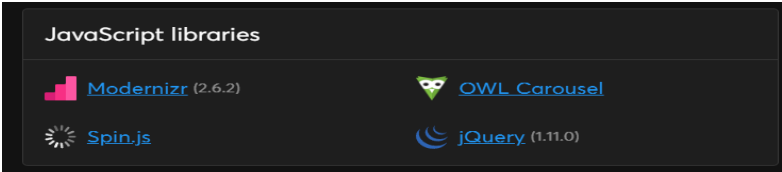


Figure 13: JavaScript libraries detected — Modernizr 2.6.2, OWL Carousel, Spin.js, and jQuery 1.11.0.

◆ CDN

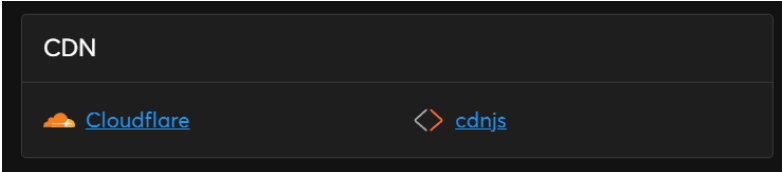


Figure 14: Content delivery split across Cloudflare and cdnjs.

◆ Technology Summary

Category	Technology Identified
Operating System	Ubuntu (Linux)
Web Server	Apache HTTP Server 2.4.41
UI Framework	Bootstrap 3.3.5

JavaScript Libraries	jQuery 1.11.0, Modernizr 2.6.2, OWL Carousel, Spin.js
CDN	Cloudflare, cdnjs
CMS / Back End	Not conclusively fingerprinted; .php extensions in robots.txt indicate a PHP-based custom back end.

Wappalyzer reveals a technology stack with notable version gaps. Bootstrap 3.3.5 was released in 2015 and is end-of-life, Apache 2.4.41 dates to 2019 and is several patch releases behind the current 2.4.x branch, and jQuery 1.11.0 (released 2014) carries known XSS vulnerabilities including CVE-2019-11358. Running outdated front-end and server-side software means the site is exposed to publicly documented CVEs that have not been patched. The CDN split between Cloudflare (primary) and cdnjs (third-party library hosting) also introduces a supply-chain dependency that should be protected with Subresource Integrity (SRI) hashes on any CDN-loaded scripts.

KEY FINDING Bootstrap 3.3.5 (2015), Apache 2.4.41 (2019), and jQuery 1.11.0 (2014) are all significantly outdated. jQuery 1.x carries known XSS CVEs. Updating these components is a recommended remediation priority.

PART E

HTTP Security Headers

HTTP response headers were analysed using SecurityHeaders.com (operated by Snyk), which grades a site’s defensive header implementation on an A+ to F scale. Security headers are instructions a web server sends to the browser to enforce policies such as preventing iframe embedding, restricting script sources, and requiring HTTPS — they are a low-effort, high-impact defensive layer.

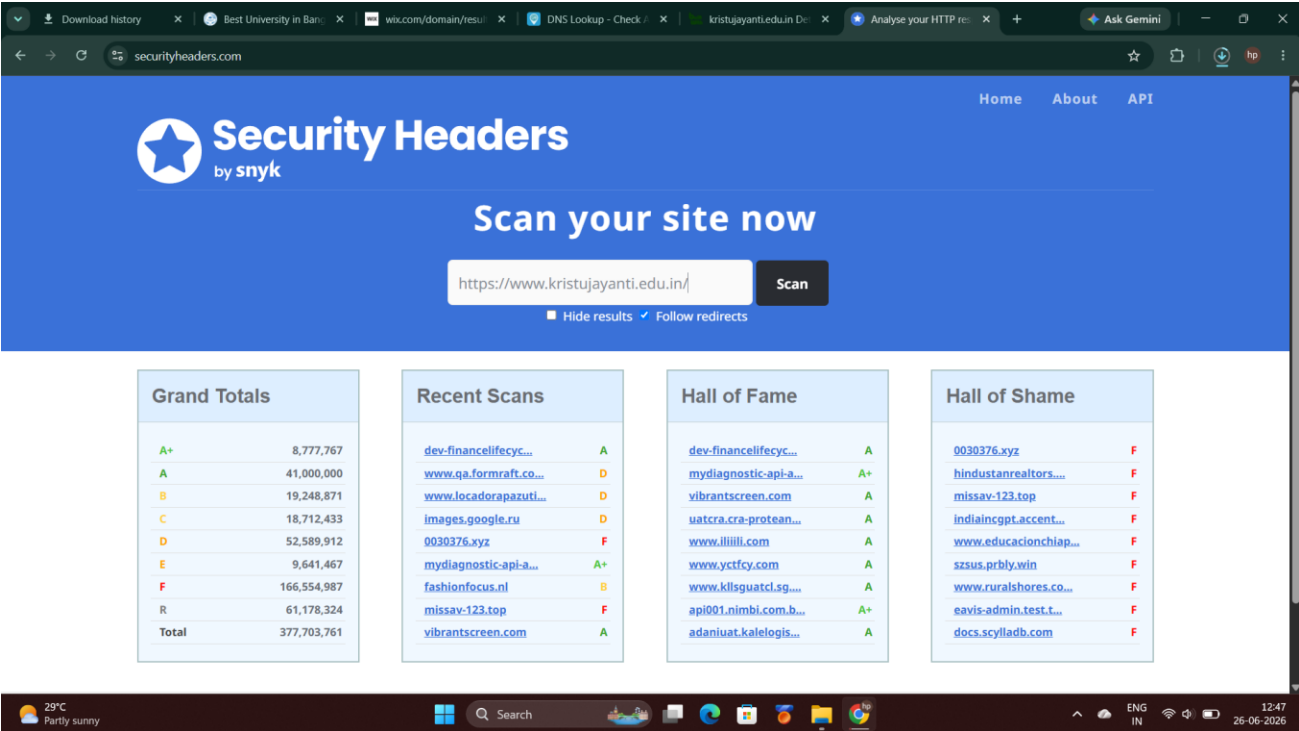


Figure 15: SecurityHeaders.com configured to analyse https://www.kristujayanti.edu.in/.

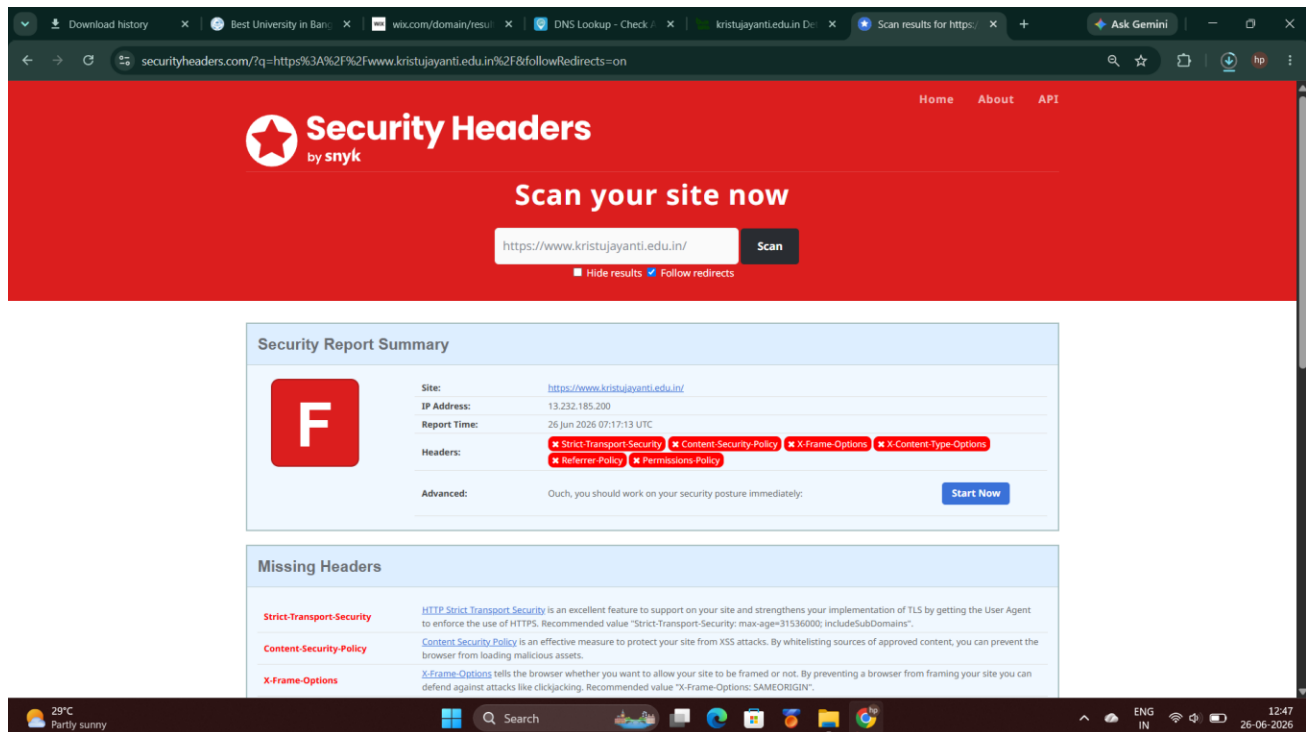


Figure 16: Scan result — overall grade of F, all six recommended security headers missing.

◆ Security Header Checklist

Header	Present	Purpose
Content-Security-Policy	No	Restricts sources from which scripts and content may load, mitigating XSS.
X-Frame-Options	No	Prevents page from being embedded in an iframe, defending against clickjacking.
X-Content-Type-Options	No	Prevents MIME-sniffing, forcing the browser to honour the declared Content-Type.
Strict-Transport-Security	No	Forces HTTPS for all future requests, preventing SSL-stripping attacks.
Referrer-Policy	No	Controls how much referrer information is shared when navigating to other sites.
Permissions-Policy	No	Restricts browser features (camera, geolocation, etc.) the page may access.

Missing Headers	
Strict-Transport-Security	HTTP Strict Transport Security is an excellent feature to support on your site and strengthens your implementation of TLS by getting the User Agent to enforce the use of HTTPS. Recommended value "Strict-Transport-Security: max-age=31536000; includeSubDomains".
Content-Security-Policy	Content Security Policy is an effective measure to protect your site from XSS attacks. By whitelisting sources of approved content, you can prevent the browser from loading malicious assets.
X-Frame-Options	X-Frame-Options tells the browser whether you want to allow your site to be framed or not. By preventing a browser from framing your site you can defend against attacks like clickjacking. Recommended value "X-Frame-Options: SAMEORIGIN".
X-Content-Type-Options	X-Content-Type-Options stops a browser from trying to MIME-sniff the content type and forces it to stick with the declared content-type. The only valid value for this header is "X-Content-Type-Options: nosniff".
Referrer-Policy	Referrer Policy is a new header that allows a site to control how much information the browser includes with navigations away from a document and should be set by all sites.
Permissions-Policy	Permissions Policy is a new header that allows a site to control which features and APIs can be used in the browser.

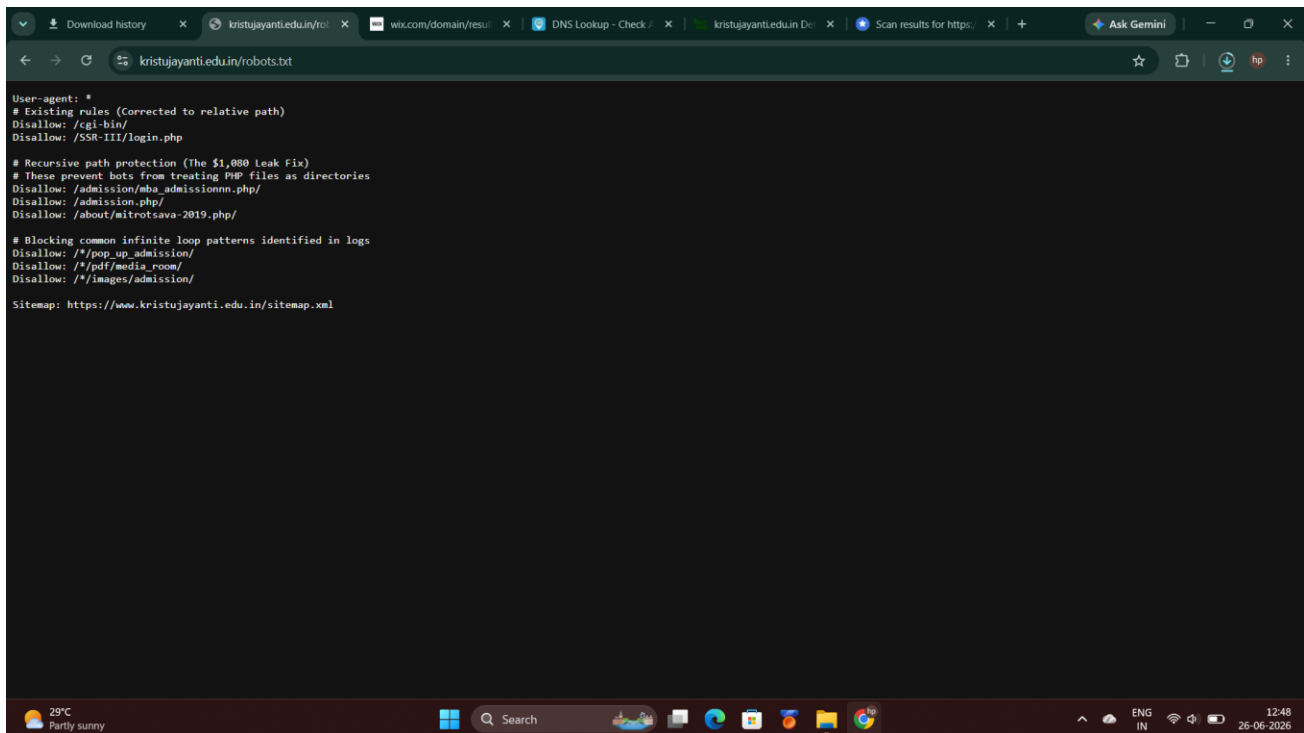
Figure 17: Detailed breakdown of all six missing security headers and their defensive purpose.

The scan returns a grade of F — the lowest possible score — because none of the six recommended security headers are present. The absence of Strict-Transport-Security means a network attacker can downgrade user connections from HTTPS to HTTP. Missing X-Frame-Options leaves the site open to clickjacking attacks where an attacker embeds the login page in a hidden iframe. Without Content-Security-Policy, any successful XSS injection can execute scripts without restriction. All six headers can typically be added in a few lines of Apache configuration or through the Cloudflare dashboard with no application code changes required.

RISK ASSESSMENT Grade: F. No security headers implemented. Recommended immediate actions: add Strict-Transport-Security, X-Content-Type-Options, and X-Frame-Options at the server or CDN layer, followed by a tailored Content-Security-Policy.

PART F**Robots.txt & Sitemap Analysis**◆ **robots.txt**

The file <https://www.kristujayanti.edu.in/robots.txt> was retrieved directly in the browser. Robots.txt is a publicly published file intended to instruct web crawlers which paths to avoid indexing — it is not an access-control mechanism and is visible to anyone.



```
User-agent: *  
# Existing rules (Corrected to relative path)  
Disallow: /cgi-bin/  
Disallow: /SSR-III/login.php  
  
# Recursive path protection (The $1,000 Leak Fix)  
# These prevent bots from treating PHP files as directories  
Disallow: /admission/aha_admissionnn.php/  
Disallow: /admission.php/  
Disallow: /about/mitrotsava-2019.php/  
  
# Blocking common infinite loop patterns identified in logs  
Disallow: /*pop_up_admission/  
Disallow: /*pdf/media_room/  
Disallow: /*images/admission/  
  
Sitemap: https://www.kristujayanti.edu.in/sitemap.xml
```

Figure 18: Contents of robots.txt, listing disallowed paths and a reference to the XML sitemap.

◆ **sitemap.xml**

The referenced sitemap was retrieved at <https://www.kristujayanti.edu.in/sitemap.xml>, generated via www.mysitemapgenerator.com.

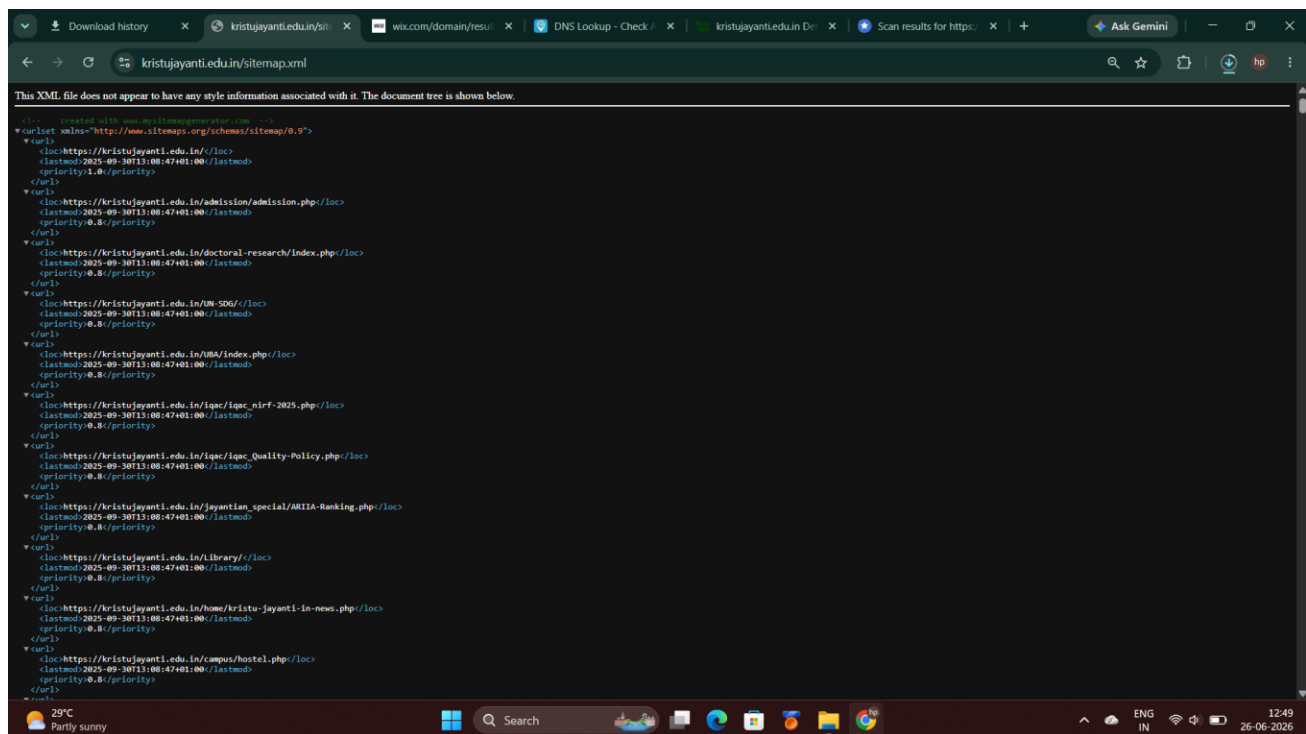


Figure 19: sitemap.xml listing indexed URLs, last-modified dates, and crawl priority values.

Together, robots.txt and sitemap.xml provide a near-complete structural map of the site without any active crawling. The robots.txt file disallows crawling of /SSR-III/login.php, directly disclosing the existence of a named internal login portal that is not linked from the public homepage. Several admission and event-related .php paths are also blocked, exposing the backend directory structure. Most notably, inline comments in the file (“Recursive path protection (The \$1,080 Leak Fix)” and “Blocking common infinite loop patterns”) reveal that these rules were added reactively after a real crawler-triggered cost incident, inadvertently documenting an operational security event in a publicly readable file. The sitemap enumerates dozens of indexed URLs and shows a static lastmod date of 30 September 2025 for every entry, indicating it was generated as a one-off snapshot rather than updated automatically.

OBSERVATION The robots.txt login path disclosure (/SSR-III/login.php) and the incident-driven comments are the most significant findings in this section: they expose internal architecture and past operational incidents to any visitor who reads the file.

PART G**Reconnaissance Report Summary**

Category	Summary
Target	kristujayanti.edu.in — Kristu Jayanti University
WHOIS	Registered via ERNET India since 2012, expiring 2035; registrant details redacted; DNS fully on AWS Route 53.
DNS	A records on Amazon Data Services India (2 IPs); mail via Amazon SES; SPF covers Google and Amazon senders.
Technology Stack	Ubuntu + Apache 2.4.41 + Bootstrap 3.3.5 + jQuery 1.11.0; CDN across Cloudflare and cdnjs; likely PHP back end.
Security Headers	Grade F — all six recommended HTTP security headers absent.
Robots.txt / Sitemap	Both present; disclose an internal login path, operational incident history, and a static URL inventory.

◆ Overall Observations

- The university has fully outsourced its infrastructure to AWS and Cloudflare, which reduces the directly managed attack surface but introduces cloud-account misconfiguration as a risk vector.
- The application layer runs significantly outdated software (Bootstrap 3.3.5, Apache 2.4.41, jQuery 1.11.0) with known CVEs, indicating that infrastructure provisioning has not been matched by application-layer patching.
- The complete absence of all six HTTP security headers (Grade F) is the most immediately actionable finding and can be remediated at the server or CDN layer with minimal effort.
- Publicly accessible files (robots.txt, sitemap.xml) inadvertently expose internal login paths, directory structure, and past operational incidents — a configuration review is recommended.

Conclusion

This exercise provided a practical introduction to the reconnaissance phase of ethical hacking, demonstrating how much information can be gathered about a target without ever touching its systems. Using only WHOIS lookups, DNS enumeration, browser-based technology fingerprinting, HTTP header analysis, and two publicly published files, it was possible to map the domain history, hosting provider, mail infrastructure, software stack, security posture, and partial site architecture of Kristu Jayanti University.

The most important takeaway is that passive reconnaissance is a synthesis exercise rather than the output of any single tool. Each data source reinforced or extended the findings of the others: WHOIS confirmed AWS name servers; DNS enumeration confirmed AWS hosting and exposed dual email channels; Wappalyzer identified outdated software; the security header scan quantified the resulting risk with a concrete grade; and robots.txt revealed internal paths and operational history that no automated scan would surface. This task reinforced why professional penetration testers invest significant time in the reconnaissance phase before active testing begins.

Topics Studied

Topic	Description
Ethical Hacking Methodology	Structured, legal approach to testing systems using attacker techniques with explicit authorisation.
Penetration Testing Phases	Reconnaissance, Scanning, Exploitation, Post-Exploitation, and Reporting.
Passive vs. Active Reconnaissance	Passive recon collects data without touching the target; active recon involves direct system interaction.
WHOIS	Protocol for querying domain registration databases for ownership, registrar, and expiry metadata.
DNS Basics	Translates domain names to IPs and stores infrastructure metadata via A, MX, NS, and TXT records.
HTTP Security Headers	Browser-enforced policies defending against XSS, clickjacking, MIME-sniffing, and SSL-stripping.
Robots.txt	Crawler instruction file that can inadvertently expose internal paths and operational history.
Sitemap.xml	Public URL inventory designed for search engines, also useful for passive site-structure mapping.
Footprinting Techniques	OSINT methods used to build a comprehensive profile of a target’s digital infrastructure.